Steam Game Analysis

- Research-Inspired Data Preparation and Programming in Python -

Research Paper Summary

Research Information

Item	Details
Title	Predicting the Popularity of Games on Steam
Authors	Andraž De Luisa, Jan Hartman, David Nabergoj, Samo Pahor, Marko Rus, Bozhidar Stevanoski, Jure Demšar, and Erik Štrumbelj
Publication year	2021
Publication	Elektrotehniški vestnik, 88(4), 151–162

The researchers wanted to understand how game attributes relate to popularity shortly after release on Steam. They used player counts from the first month to predict player activity in the second month. They also examined whether price, supported languages, storage requirements, release date, and game genres could improve the predictions. The study aimed to help developers understand early game performance. It considered both the predicted player count and the uncertainty around that prediction.

Dataset Description (Original)

The researchers collected data from Steam, SteamSpy, and SteamDB over one week in mid-December 2020. They focused on games released after 2015.

Item	Description
Number of records	8,000 games used in the experiments
Main numerical features	First-month player count, price, number of supported languages, and storage requirements
Target variable	Median player count in the second month after release
Time features	Release year, day of the year, and day of the month
Categorical information	Game genres, including 33 genre categories

The paper describes 4 main numerical features, 3 time features, and genre information. Popularity in this study means player activity. It does not directly measure sales, revenue, or user satisfaction.

Data Preparation Techniques

Aspect	Method
Missing values	No complete handling procedure reported.
Categorical variables	Release year treated as categorical; genres used in hierarchical models.
Normalization and scaling	Prices converted to EUR; storage to MB; numerical features log-transformed; dates scaled and cyclically encoded.
Feature selection	Retained player counts, price, languages, storage, dates, and genres; excluded developers and publishers due to many unique categories.
Data splitting	No fixed train/test split reported; evaluated using PSIS-LOOIC, an approximation of leave-one-out cross-validation.

Main Findings and Relevance

First-month player activity was the strongest predictor of second-month activity. A model that accounted for game genres and allowed prediction uncertainty to vary between games performed best. Predictions became less accurate when looking further into the future.

For our project, the main lesson is to define the analysis measure clearly and prepare each field according to its meaning. We use these ideas for data preparation and review analysis with a different public dataset. Our results therefore describe review activity and do not reproduce the paper's player-count predictions.

Project Data and Python Workflow

Dataset Information

We used the public *Steam Games Dataset* maintained by FronkonGames on Hugging Face. It contains game information collected from Steam and SteamSpy.

Total reviews describe review activity, while positive review rate describes reviewer sentiment. The data is a fixed snapshot and does not show changes over time.

Item	Details
File	games.csv
File size	401.1 MB
Original records	125,855
Original fields	40 after correcting the header
Selected release period	2020–2025
Minimum reviews	10 per game
Final sample	32,387 games

Python Workflow

Step	Description
Download	Saved a fixed CSV version from Hugging Face locally.
Inspect	Checked fields, data types, missing values, duplicate IDs, and dates.
Correct the header	Separated DiscountDLC count into two columns.
Select fields	Kept 14 fields relevant to the analysis.
Filter dates	Selected games released from 2020 to 2025.
Convert text	Converted languages, genres, and categories into lists.
Add measures	Calculated total reviews and positive review rates.
Filter reviews	Kept games with at least 10 reviews.
Aggregate	Calculated yearly counts, means, and medians.

Record Selection

Stage	Records retained
Original dataset	125,855
After filtering release dates	91,763
After filtering review counts	32,387

The review threshold reduces the influence of games with very few reviews. Excluded games are outside our study scope, this does not mean that nobody played them.

Data Quality Issues and Handling

Issue	Handling (Step in Notebook)
Header mismatch	Split DiscountDLC count into Discount and DLC count, then reload the CSV. (Step 3)
Missing genres and categories	Return [ ] for missing values in split_items(). (Step 5)
Irregular language quotation marks	Remove outer brackets and surrounding quotes, then split by commas. (Step 5)
Wide numerical ranges	Plot log(1 + reviews) and log(1 + price); summarize reviews using both means and medians. (Step 7)
Ambiguous activity zeros	Calculate correlations on all records and separately on records where each activity value is greater than zero. (Step 6, 7)

The original CSV was left unchanged. Empty lists indicate unavailable information, rather than confirming that a game has no genres or categories.

Activity field	Zero values*	Percentage
Average playtime	22,921	70.8%
Peak CCU (Concurrent Users)	21,621	66.8%

** Zero value doesn't mean that nobody played them.*

Visualizations and Findings

Games by Release Year

Figure 1 below compares game counts by release year. In our sample, 2024 has the most games (6,889), while 2025 has the fewest (3,156). This reflects the selected data rather than all Steam releases; dataset coverage and time to collect reviews may explain some of the difference.

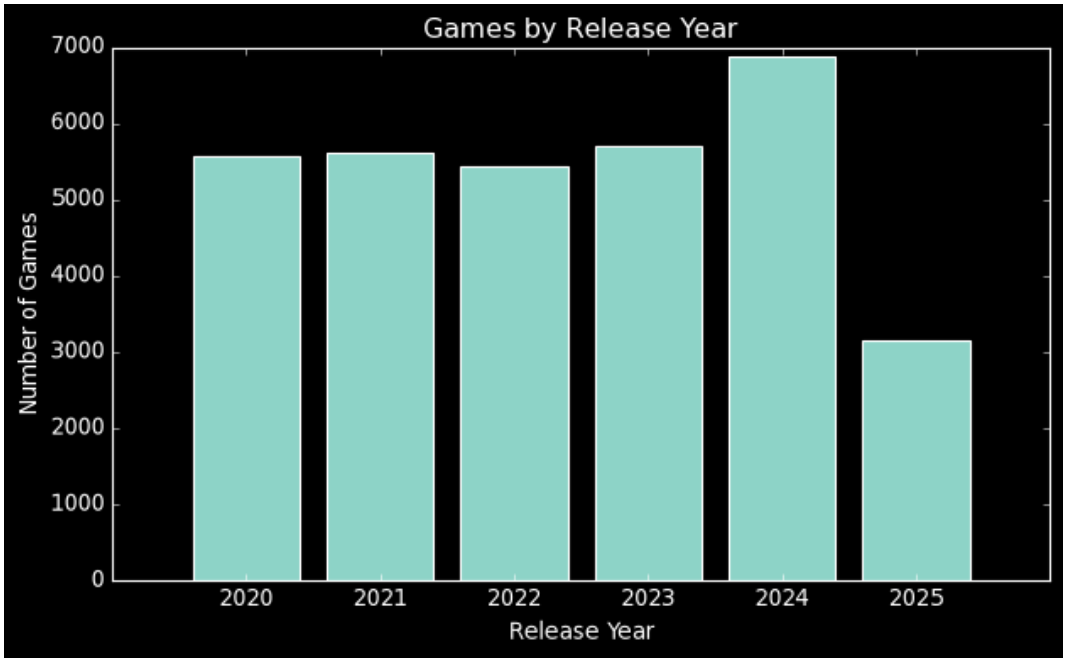


Figure 1. Selected games by release year

Distribution of Review Counts

Figure 2 below shows that most games have relatively few reviews, while a small number have much higher counts. We used log(1 + reviews) to make this wide range easier to see. The long right tail helps explain why mean review counts are much higher than medians.

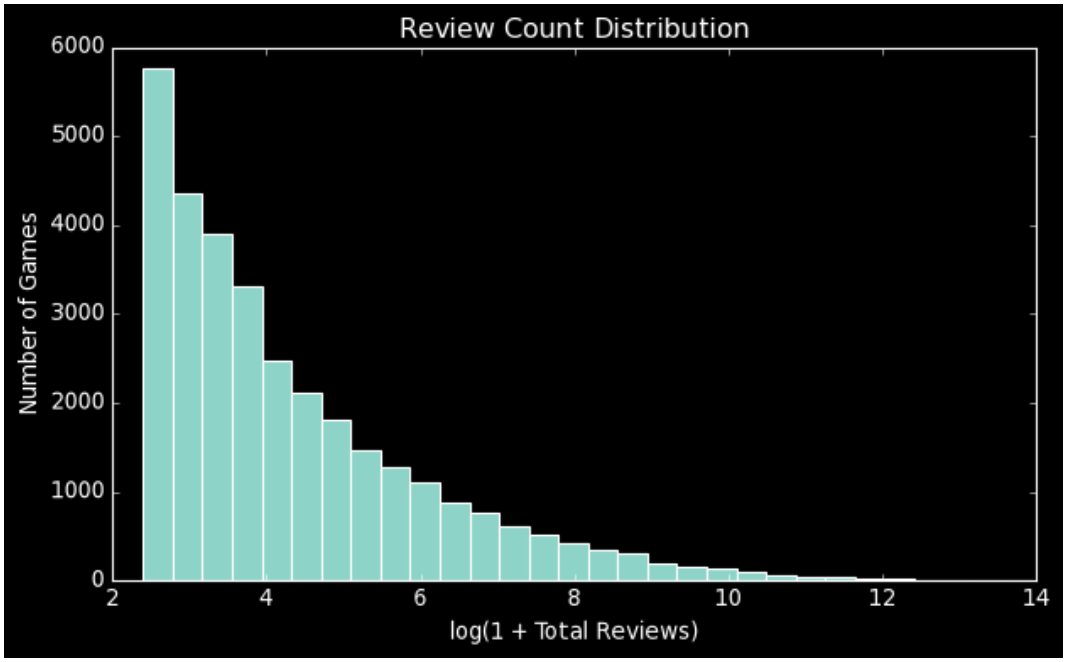


Figure 2. Distribution of log-transformed review counts

Price and Review Activity

Figure 3 below compares prices and review counts using log transformations. Games at similar prices can have very different review counts. Zero-priced games also show a wide range of review activity, suggesting that price alone does not explain the differences.

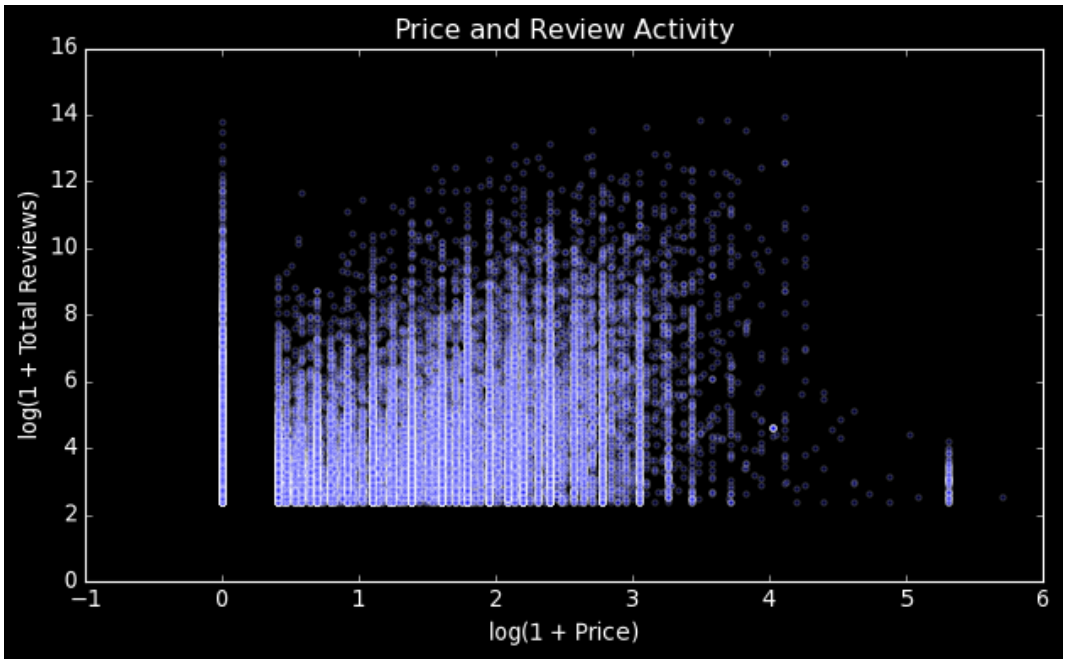


Figure 3. Game prices and review activity

Correlation with Review Activity

Using Spearman correlation to compare game attributes with review counts. Positive values mean that higher values of one variable generally go with higher values of the other.

Feature	Correlation with review counts
Average playtime	0.713
Peak CCU	0.617
Language count	0.323
Price	0.235

Average playtime shows the strongest association, followed by the dataset's CCU field. Language count and price have weaker positive associations, while release year shows almost no association.

Effect of Recorded Zeros

Feature	All records	Positive values only	Records in positive subset
Average playtime	0.713	0.500	9,466
Peak CCU	0.617	0.670	10,766

The results change when zeros are excluded. Both activity measures remain positively associated with reviews, but their relative strength depends on which records are included. The two positive subsets contain different games, so they cannot provide a direct ranking of importance. These results show associations, not causal effects.

Correlation Findings

Average playtime and Peak CCU were most strongly associated with review counts (0.713 and 0.617). Excluding zeros changed these values to 0.500 and 0.670, using different subsets. The results depend on data coverage and do not show causal effects.

Conclusion

What I Learned

This project helped me practice loading, cleaning, filtering, and summarizing data with Python. I learned to check the CSV structure before analysis and to handle text fields using functions and string methods. The paper also helped me understand why clear measures matter. The zero-value comparison showed how data choices can change results, so correlations need careful interpretation.

References

1. De Luisa, A., et al. (2021). *Predicting the Popularity of Games on Steam*. Elektrotehniški vestnik, 88(4), 151–162. Paper
2. FronkonGames. *Steam Games Dataset*. Hugging Face. Revision: 52ba62abfa707a2fa31f4d147c2777a9d9fbd2da. Dataset
3. FronkonGames. *Steam Games Dataset Documentation*. Dataset card
4. FronkonGames. *Steam Games Scraper*. Collection code and documentation